$$h_i = \text{Attention}(XW_i^Q, XW_i^K, XW_i^V),$$

$$\text{MultiHead}(X) = \text{Concat}(h_1, \ldots, h_8)W^O,$$

$$W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{512\times 64}, \quad W^O \in \mathbb{R}^{512\times 512}$$